Higher-order Repetitive Behaviors in Toddlers Born Preterm

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Background

- Preterm birth is associated with lower IQ, impaired self-regulation, and social communication deficits.
- Preterm infants are also more likely to receive psychiatric diagnoses including ASD, the prevalence of which is 7% in children born preterm (as compared to 1.5% in the general population)[1].
- These findings add to the extant literature suggesting the **heterogeneous etiology of ASD**, and highlight the need for screening tools that **capture the dimensionality and heterogeneity** in contributing mechanisms and clinical outcomes of ASD.
- Though Restricted and Repetitive Behaviors (RRBs) have been studied extensively in ASD populations, less is known about dimensional differences in RRBs in toddlers born preterm.

Aims

Aim1: Measure dimensional features of Restricted and Repetitive Behaviors in a large, community sample.

Aim 2: Compare Restricted and Repetitive Behaviors in toddlers born preterm to fullterm controls.

Aim 3: Determine how group differences are related to Recipal Social Behaviors in this Sample.

Methods

Participants. Parents of toddlers between 17 and 23 months were recruited from a community sample (n = 1,995).

Exclusionary Criteria:

- Failure to complete both measures (384 excluded).
- Gestation lengths classified as Extremely preterm (< 28 weeks gestation length) or Post-term (>42 weeks gestation length) (22 excluded).

	Age	Gestation length (Mean, SE)	N	Males	Females
Preterm	20.4 (.022)	34.8, .017	104	60	44
Fullterm	20.0 (.001)	39.72, .001	1485	774	711
			1589	834	755

Table 1. Final sample of toddlers (n=1,589)

Measures

1. 34-item Repetitive Behavior Scale for Early Childhood (RBS-EC), a parent report that captures dimensional features across a broad range of restricted and repetitive behaviors.
2. 48-item Video-Referenced Rating of Reciprocal Social Behaviors (vr-RSB), a parent report that captures quantitative aspects of reciprocal social behavior.

Results

Aim1: Measure dimensional features of Restricted and Repetitive Behaviors in a large, community sample.

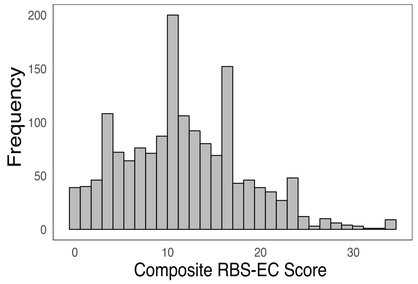


Figure 1. The average RBS-EC coposite topography score for the sample was 12 (SD = 6.6), replicating Wolff, Boyd & Elison (2016)[2].

Aim 2: Compare Restricted and Repetitive Behaviors in toddlers born preterm to fullterm controls.

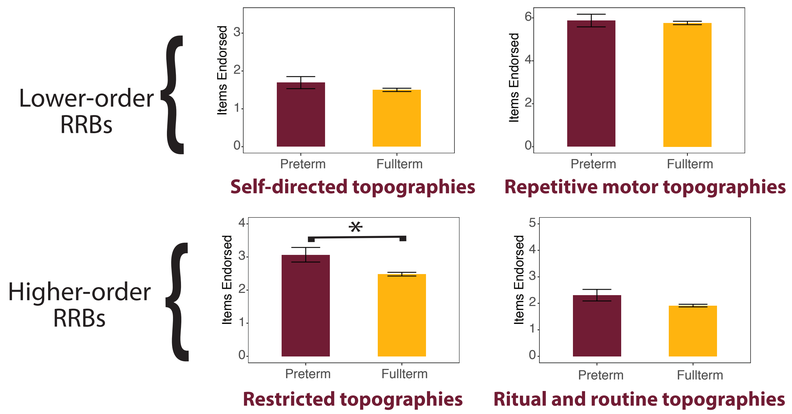
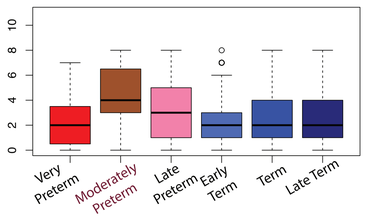


Figure 2. Bar plots show RRBs broken down by topographies. Parents of toddlers born preterm reported significantly more items designed as "Restricted" than toddlers born fullterm (t=2.6, p<.05, p-values adjusted for multiple comparisons).

Figure 3. Restricted topographies broken down by clinically-defined gestation groups. Post-hoc paired-sample t-tests revealed that parents of toddlers born Moderately Preterm reported significantly greater Restricted Topography items than toddlers born Early Term, Term, and Late Term (all p's < .05, adjusted using Tukey's HSD).



Aim 3: Determine how group differences are related to Reciprocal Social Behaviors in this Sample.

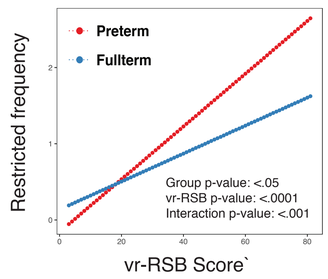


Figure 4. There was a significant interaction between Group and vr-RSB in predicting the Frequency of Restricted Topography items. vr-RSB Scores were more strongly associated with Restricted Topography items in Toddlers born Preterm than in Toddlers Born Fullterm (F(1585) = 48.7, r^2 = .08, p<.0001).

Conclusion

- RBS-EC measures collected from a large community-based sample (n=1,589, 104 preterm) indicate that RRBs are normally distributed among toddlers, replicating findings from Wolff, Boyd & Elison (2016)[2].
- Toddlers born preterm demonstrated a higher degree of RRBs on the **Restricted topography** subscale.
  - This effect was driven primarily by the Moderately Preterm Group (gestation length = 32-33 weeks).
- **Reciprocal Social Behaviors** were associated with the Frequency of Occurrence of Restricted Items. This effect **interacted with Preterm Birth**, such that vr-RSB scores were more predictive of the Frequency of Restricted Items for toddlers born preterm.

Discussion and Future Directions

- The present study has important implications for understnading RRBs, a key diagnostic features of ASD. These findings suggest that the RBS-EC can be used to examine:
  1. The dimensionality of sub-clinical RRBs.
  2. Differences in Restricted and Repetitive Behaviors in toddlers born preterm.
  3. How Restricted and Repetitive Behaviors relate to Reciprocal Social Behaviors.
- **Future analyses** will:
  - Use dimensional measures of gestation length and birthweight to model the relationship between preterm birth and Restricted and Repetitive Behaviors.
  - Work towards undersanding the non-linear relationship between gestation length and Restricted topographies (**Figure 3**).

References

1. Johnson, S., Hollis, C., Kocchar, P., Hennessy, E., Wolke, D., & Marlow, N. (2010). Autism spectrum disorders in extremely preterm children. Journal of Pediatrics.
2. Wolff, J. J., Boyd, B. A., & Elison, J. T. (2016). A quantitative measure of restricted and repetitive behaviors for early childhood. Journal of Neurodevelopmental Disorders, 1–10.

Contact Information

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